

Physics 4730, Fall 2026: Lab 09

Please complete the calculations below in class. We will make use of the results in Lab 10.

1. The discrete Poisson probability density function is given by:

$$f(x|\lambda) = \frac{\lambda^x}{x!} e^{-\lambda}$$

where x is a nonnegative integer. Write down an expression for the *likelihood function* L for a set of measurements x_i of a Poisson random variable. Apply any simplifications you can think of.

2. Now, find the *log-likelihood* function $\ln(L)$. Apply identities to break it into three terms: one dependent on x_i , one dependent on $x_i!$, and a constant.

3. Finally, by maximizing $\ln(L)$ with respect to λ , find an expression for the MLE for λ .